厦门兰阶教育 2022 年学科奥赛•物理珍题集训

模拟试卷二（卷Ⅰ）　命题者：苏国珍教授

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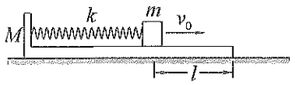
一.（36 分）如图，质量为 M 的长木板静置于光滑水平地面上，劲度系数为 k 的轻弹簧一端固定在木块的左端，另一端与质量为 m 的物块（可视为质点）相连，物块置于木板上，它与木板间的滑动摩擦系数（等于静摩擦系数）为 μ_k。开始时，木板与物块均保持静止，弹簧无形变，物块与木板右端的距离为 l。今给物块一水平向右的冲量，使其瞬间获得大小为 v_0 的速度。设 $M=2m$，$k=4mg/l$，$\mu_k=1/4$，重力加速度为 g。

（1）为使物块不会脱离木板，v_0 应满足什么条件？[8 分]

（2）由于物块与木板间存在摩擦力，一段时间后，物块相对于木板的运动将停止。假设 v_0 取满足上述条件的上限，试求物块相对于木板的运动最终停止时，物块在木板上的位置（用物块到木板右端的距离表示）。[16 分]

（3）求从物块开始运动到物块相对木板停止运动，木板相对于地面的位移。[12 分]



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二.（40 分）类星体（Quasar）是极度明亮的活动星系核。大多数星系的核心都有一个特大质量的黑洞，它的质量从百万至数十亿太阳质量不等。在类星体和其它形式的活动星系核，黑洞被气态的吸积盘环绕着。由于中心星体自转改变吸积盘表面的磁场的原因，吸积盘中的气体朝向黑洞坠落，能量就会以电磁辐射和物质喷流的形式释放出来。下图为类星体的一个示意图，其吸积盘和喷流均清晰可见。

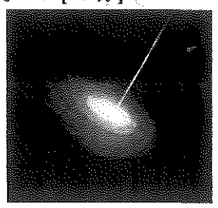
为了方便计算，本题中我们考虑一个简化的等效模型：设类星体中心为半径可略的质量 M 的重星系核，吸积盘等效为一质量面密度 σ，半径 r，以星系核为圆心的匀质圆盘，而喷流则等效为质量线密度 λ 的垂直于吸积盘面的无限长匀质链条，且不考虑相对论效应。

有一质量为 m 的探测器在体系万有引力作用下，在吸积盘所在平面上运动，其轨道可近似为以星系核为焦点，离心率 e 很小的一个椭圆，其半长轴为 R。采取合理近似：$r \ll R$，$\sigma r^2 \ll M$，$\lambda R \ll M$。

（1）求当探测器距离类星体中心为 R 时，探测器受到的万有引力（吸积盘不能视为质点）；[16 分]

（2）假设探测器的运动轨道可近似为半径为 R 的圆轨道，求探测器沿轨道运行的角速度；[8 分]

（3）由于体系对探测器万有引力对平方反比律的偏移，探测器在每一个周期内距离星系核最近的点（近心点）将依次转过一个角度，可以认为整个轨道以一个小的角速度 Ω 绕星系核转动。 计算这个转动角速度 Ω。[16 分]



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三.（46 分）如图 1，一质量为 m，底半径为 R 的匀质圆柱体，被挖去一小圆柱区域。挖去部分的半径为 $R/2$，侧面与大圆柱体的侧面相切。

（1）求柱体（指余下的柱体，下同）的质心 C 到原圆柱体轴线 OO' 的距离 b 以及柱体绕过质心 C 且与 OO' 平行的轴线的转动惯量 J_C。[6 分]

（2）将柱体侧放在一个倾角为 θ 的粗糙斜面上。用质心 C 到轴线 OO' 的垂线 OC 与铅直线的夹角 φ 表示柱体在斜面上的角位置，如图 2 所示。

（2.1）若柱体在斜面上能处于静态平衡，求斜面倾角 θ 和柱体与斜面之间的静摩擦系数 μ 所需满足的条件；[6 分]

（2.2）在满足上述条件下，求平衡位置所对应 φ 的大小（记为 φ_0），并判断平衡的稳定性；[6 分]

（2.3）若柱体在斜面上存在稳定平衡位置，求其在该位置附近作微振动的角频率 ω，结果可用 R、θ、φ_0 和重力加速度 g 表示。[12 分]

（2.4）假设柱体能围绕平衡位置作振幅（角位置 φ 偏离 φ_0 的最大值）为 δ_0 的微振动，且在振动过程中柱体与斜面之间无相对滑动，则斜面倾角 θ 和柱体与斜面之间的静摩擦系数 μ 应满足什么条件？θ 满足的条件要求精确给出，μ 满足的条件可用 θ、φ_0 和 δ_0 表示，结果只需保留到 δ_0 的一次方项。[16 分]

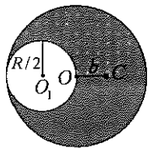

图1

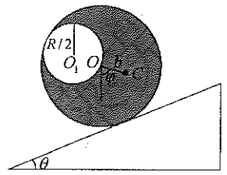

图2

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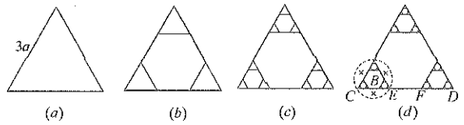
四.（40 分）用单位长度电阻为 λ 的均匀电阻丝连成边长为 $3a$ 的等边三角形，如图(a)；再用相同的电阻丝连接三边的三等分点，形成图(b)；再连接三个小三角形的三等分点，形成图(c)；以此类推，操作无限多次，最终形成图(d)。

（1）求图(d)中 C、D 两点之间的电阻。[10 分]

（2）在图(d)左下三角形的外接圆区域有垂直于三角形平面，大小随时间均匀增大的匀强磁场 $B=kt$（k 为常量）。求图(d)中流经 EF 段的电流。[18 分]

（3）续第（2）问，求图(d)中 E、F 两点之间的电势差。[12 分]



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五.（42 分）角向箍缩利用电磁场加速等离子体，使之完成对靶棒的向心聚爆，其装置如图所示。在作为真空管的圆形玻璃管外围了一匝导体板，真空管内含一段同长度的圆圆柱靶棒。已知玻璃管的长度为 H，底半径为 b，且 $H>>b$，靶棒的截面半径为 a，导体板由厚度为 d，电导率为 γ 的均匀导电材料制成，电源的电压为恒定值 V，管内充有氢气，且已电离为正负离子数密度均为 n 的等离子体。假设玻璃管管壁厚度和导体板厚度均远小于玻璃管底半径 b，除了导体板外，其余部分（包括电极、导线和电源内部）的电阻均可忽略不计，电路中各部分电容均可忽略不计，靶棒和玻璃管均为非磁性材料（磁导率近似为 μ_0），除了玻璃管的内部（含靶棒内部）外，其余部分的磁通均可忽略不计。在 $t=0$ 时刻接通电源。

（1）求通过电源的电流 i 与时间 t 的关系。[8 分]

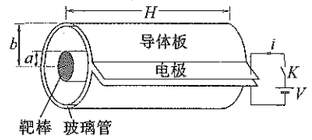
（2）假设电路中的电流在极短时间内增大到稳定值 i_s，之后电流保持为 i_s 不变，不计粒子的热运动、库仑相互作用以及粒子间的碰撞，记电子电量为 $-e$，电子和氢核的质量分别为 m_e 和 m_p。

（2.1）求电流刚达到 i_s 的瞬间，距中心轴 $r\ (a<r<b)$ 处带电量为 q，质量为 m 的粒子的速度 v_0；[8 分]

（2.2）求上一问中的粒子运动到 $r=a$ 处的时间 $t(r)$；[8 分]

（2.3）假设离子与靶棒发生完全非弹性碰撞，求 t 时刻靶棒表面的压强 $P(t)$；[10 分]

（2.4）记靶棒表面可达到的最大压强为 $P_{\max}$，并定义 $\beta=P_{\max}/w_B$，式中 $w_B=B^2/(2\mu_0)$ 为真空管内磁场的能量密度。取 $\mu_0=4\pi\times10^{-7}\,\mathrm{N/A^2}$，$H=30.0\,\mathrm{m}$，$a=1.0\,\mathrm{mm}$，$b=1.00\,\mathrm{m}$，$n=1.00\times10^{8}\,\mathrm{m^{-3}}$，$m_e=9.10\times10^{-31}\,\mathrm{kg}$，$e=1.6\times10^{-19}\,\mathrm{C}$。计算此装置的 β 值，再与绝大多数托卡马克装置的 β ($\approx0.1\sim1$) 进行对比，并用简洁的文字表述角向箍缩的优点。[8 分]



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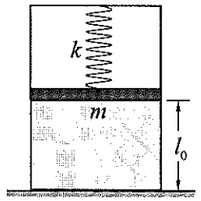
六.（38 分）如图所示，一圆柱体气缸置于水平地面上。质量为 m 的水平放置的活塞可以在气缸中无摩擦地上下运动，活塞被一无质量的弹簧与气缸上端连接。活塞上方为真空，下方含有一定量的理想气体氦气。开始阶段活塞与氦气处于平衡状态时，弹簧处于未形变状态，此时活塞到汽缸底部的距离为 l_0。设弹簧的劲度系数 $k = mg / l_0$。忽略气体渗漏，气缸、活塞都是绝热的，与活塞质量 m 相比，气体的质量可忽略。

（1）活塞因受微小扰动而上下振动，计算其振动周期。[10 分]

（2）从初态出发将活塞向下压缩气体至原来体积的一半，无初速度释放活塞任其运动。

（2.1）设汽缸的质量足够大，求活塞能升起的最大高度。[12 分]

（2.2）设汽缸的质量可忽略，讨论活塞上升过程中汽缸从地面被提起的可能性；若能被提起，求出能被提离地面的最大高度；若不能被提起，请说明理由。[16 分]



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七.（38 分）激光雷达是利用激光束进行目标探测的雷达系统，工作原理是向目标发射探测信号（激光束），接收目标反射的信号（目标回波），并将其与发射信号进行比较，经分析处理即可获取目标的距离、速度等信息。激光雷达具有分辨率高、隐蔽性好、抗干扰能力强等优点，在自动驾驶等领域应用前景广阔。

某激光雷达如图 1 所示，入射激光经分束器后一束射向待测物体（如汽车），另一束直接射向接收器。射向物体的光被物体反射后沿原路返回，经分束器到达接收器。调节分束器使得两束激光到达接收器时的振幅相同，接收器接收二者叠加后的信号。入射激光可视为简谐波，但其频率 $f_{入}$ 并非定值，而是在频率范围 $f_0 \pm B/2$ 内按周期为 T 的三角波变化，如图 2 所示，其中 $B << f_0$，$T >> 1/f_0$。

（1）用此激光雷达探测某静止物体 $(v=0)$，发现接收器收到的信号强度随时间变化。若用 l 表示物体与分束器的距离，用 f_{amp} 表示信号强度变化的频率 $(1/T << f_{amp} << f_0)$，试在 $2l/c \le t \le T+2l/c$（c 为光速）范围内画出 f_{amp} 与时间 t 关系曲线的示意图，标出关键点坐标。若实验测得 f_{amp} 的最大值为 f_m，试求 l 的表达式（结果用 f_m、B、T 和 c 表示）。[20 分]

（2）用此激光雷达探测运动的物体，发现收到的信号强度的变化频率 f_{amp} 与时间 t 的关系与物体静止时有所不同。设待测物体的速度大小为 v，速度方向与反射光线的夹角为 α，物体与分束器的距离为 l，且在观测时间内上述各量可近似认为是不变的，物体速度远小于光速。试在 $2l/c \le t \le T+2l/c$（c 为真空中光速）范围内画出 f_{amp} 与时间 t 关系曲线的示意图，标出关键点坐标。若实验测得此时 f_{amp} 的最大值为 f'_m，试求物体速度大小 v 的表达式（结果用 f'_m、f_0、B、T、l、α 和 c 表示）。[18 分]

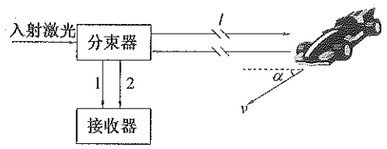

图1

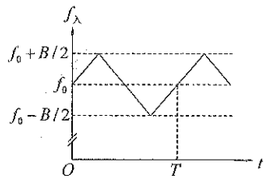

图2

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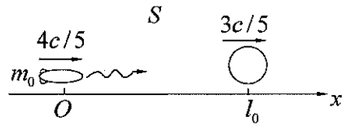
八.（40 分）如图，宇宙飞船的静质量（含燃料）为m_0，在惯性系S中以$4c/5$（c为真空中的光速）的速度沿Ox的方向飞行，正前方有一小行星正以$3c/5$的速度与宇宙飞船同向运行。为了避免与小行星发生碰撞，当宇宙飞船与小行星相距为l_0时，宇宙飞船开始向正前方发射光子。在宇宙飞船中测，单位时间发射的光子能量恒为w。试求：

（1）在宇宙飞船中测，刚开始发射光子的时刻，宇宙飞船与小行星的距离；[10 分]

（2）在惯性系S中测，刚开始发射光子的时刻，宇宙飞船的加速度；[10 分]

（3）为了使宇宙飞船不会与小行星相碰，w至少为多少？假设小行星的速度始终保持不变，宇宙飞船携带有足够的燃料。[20 分]



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中学生物理奥赛培训

模拟试卷（二）解答

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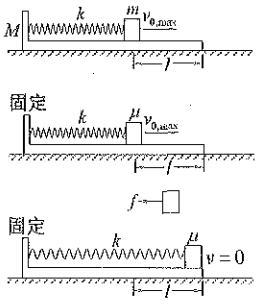
福建 厦门 2022.6

题一.

(1) $\mu=\dfrac{mM}{m+M}=\dfrac{2}{3}m$



$$\begin{cases}-f\,l=\dfrac{1}{2}kl^2-\dfrac{1}{2}\mu v_{0,\max}^2\\ f=\mu_k mg,\ \mu=2m/3\end{cases}$$

$$-\mu_k mg\,l=\frac{1}{2}kl^2-\frac{1}{3}mv_{0,\max}^2$$

$$v_{0,\max}=\sqrt{\frac{3kl^2}{2m}+3\mu_k g\,l}=\frac{3}{2}\sqrt{3gl}$$

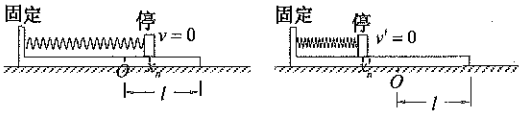
$$v_0\le\frac{3}{2}\sqrt{3gl}$$

(2)

分析：

- 物块将来回运动；
- 物块停止的位置：瞬时速度为零的位置，即来回运动最左端或最右端；
- 物块最终停下来的条件：在最左端或最右端，弹力小于最大静摩擦力。

$x_n=?\ x_n'=?$



考察物体第n次来回运动：$x_n\to x_n'\to x_{n+1}$

$$x_n\to x_n':\ -\mu_k mg(x_n-x_n')=\frac{1}{2}kx_n'^2-\frac{1}{2}kx_n^2$$

$$x_n'\to x_{n+1}:\ -\mu m_k g(x_{n+1}-x_n')=\frac{1}{2}kx_{n+1}^2-\frac{1}{2}kx_n'^2$$

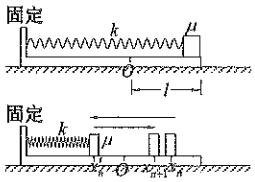
$$\begin{cases}kx_n+kx_n'=2\mu_k mg\\ kx_n'+kx_{n+1}=-2\mu_k mg\end{cases}$$

$$kx_{n+1}=kx_n-4\mu_k mg$$

$$x_1=l$$

$$kx_n=kl-(4n-4)\mu_k mg$$

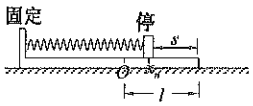
$$kx_n'=2\mu mg-kx_n=-kl+(4n-2)\mu_k mg$$



设物体停止在位置x_n：

$$\begin{cases}kx_n=kl-(4n-4)\mu_k mg<\mu_k mg\\ k=4mg/l,\ \mu_k=1/4\end{cases}$$

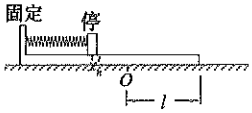
$n>4.75$，物体停止在x_4



设物体停止在位置x'_n：

$$\begin{cases}-kx_n'=kl-(4n-2)\mu_k mg<\mu_k mg\\ k=4mg/l,\ \mu_k=1/4\end{cases}$$

$n>4.25$，物体停止在x'_4



物体最终停止的位置为：

$$x_4=l-(4\times4-4)\frac{\mu_k mg}{k}=\frac{1}{4}l\qquad s=l-\frac{1}{4}l=\frac{3}{4}l$$

$$kx_n=kl-(4n-4)\mu_k mg,\ kx_n'=-kl+(4n-2)\mu_k mg$$

(3)

物块来回一次的时间：

$$T=2\pi\sqrt{\frac{\mu}{k}}=2\pi\sqrt{\frac{l}{6g}}$$

从物块开始运动到最终停止所经历的时间：

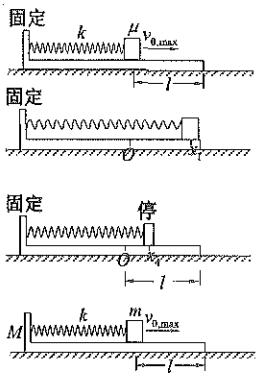
$$\Delta t=(\frac{1}{2}+3)T=7\pi\sqrt{\frac{l}{6g}}$$

质心的速度：

$$v_C=\frac{mv_{0,\max}}{m+M}=\frac{1}{3}v_{0,\max}=\frac{1}{2}\sqrt{3gl}$$

质心的位移：

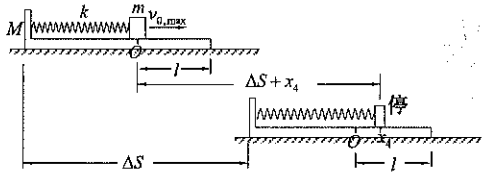
$$\Delta x_C=v_C\Delta t=\frac{7\sqrt{2}\pi}{4}l$$



$$\mu=\frac{2}{3}m,\ v_{0,\max}=\frac{3}{2}\sqrt{3gl}$$

$$\Delta x_C = \frac{m(\Delta S + x_4) + M\Delta S}{m+M}$$

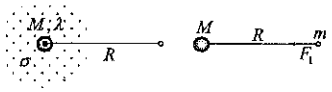
$$\Delta S = \Delta x_C - \frac{m x_4}{m+M} = (\frac{7\sqrt{2}\pi}{4} - \frac{1}{12})l$$



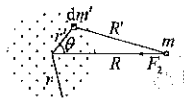
$$\Delta x_C = \frac{7\sqrt{2}\pi}{4} l, x_4 = \frac{1}{4} l$$

题二.

(1) $E_{p_1} = -\frac{GmM}{R}$



$$F_1 = -\frac{dE_{p_1}}{dR} = -\frac{GmM}{R^2}$$



$$dE_{p2} = \frac{Gmdm'}{R'} = \frac{Gm\sigma r' dr' d\theta}{\sqrt{R^2 + r'^2 - 2Rr'\cos\theta}}$$

$$\approx -\frac{Gm\sigma}{R}[1 + \frac{r'}{R}\cos\theta + (3\cos^2\theta - 1)\frac{r'^2}{2R^2}]r'dr'd\theta$$

$$= -\frac{Gm\sigma}{R}\int_0^r [\int_0^{2\pi} d\theta + \frac{r'}{R}\int_0^{2\pi}\cos\theta + \frac{r'^2}{2R^2}\int_0^{2\pi}(3\cos^2\theta - 1)]d\theta]r'dr'$$

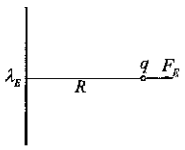
$$= -\frac{Gm\pi\sigma}{R}\int_0^r (2 + \frac{r'^2}{2R^2})r'dr' = -\frac{Gm\sigma\pi r^2}{R} - \frac{Gm\sigma\pi r^4}{8R^3}$$

$$F_2 = -\frac{dE_{p_2}}{dR} = -\frac{Gm\sigma\pi r^2}{R^2} - \frac{3Gm\sigma\pi r^4}{8R^4}$$

$$F_E = qE = \frac{q\lambda_E}{2\pi\varepsilon_0 R}$$

$$F_G = -G\frac{mM}{r^2}, F_E = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^2}$$

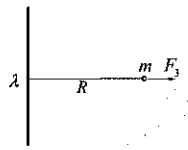
$$q \leftrightarrow -m,\ Q \leftrightarrow M,\ \frac{1}{4\pi\varepsilon_0} \leftrightarrow G,\ \lambda_E \leftrightarrow \lambda$$



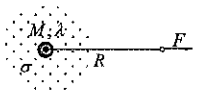
$$F_3 = -\frac{2Gm\lambda}{R}$$

$$F = F_1 + F_2 + F_3$$

$$= -\frac{GmM}{R^2}\left(1 + \frac{\sigma\pi r^2}{M} + \frac{3\sigma\pi r^4}{8MR^2} + \frac{2\lambda R}{M}\right)$$



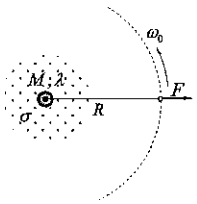
$$F_1 = -\frac{GmM}{R^2}, F_2 = -\frac{Gm\sigma\pi r^2}{R^2} - \frac{3Gm\sigma\pi r^4}{8R^4}$$



(2)

$$F_{eff}(\rho) = F(\rho) + \frac{L^2}{m\rho^3}$$

$$F_{eff}(R) = F(R) + \frac{L^2}{mR^3} = 0$$



$$L = \sqrt{-mR^3 F(R)}$$

$$= m\sqrt{GMR(1 + \frac{\sigma\pi r^2}{M} + \frac{3\sigma\pi r^4}{8MR^2} + \frac{2\lambda R}{M})} = mR^2\omega_0$$

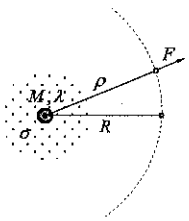
$$\omega_0 = \sqrt{\frac{GM}{R^3}(1 + \frac{\sigma\pi r^2}{M} + \frac{3\sigma\pi r^4}{8MR^2} + \frac{2\lambda R}{M})}$$

$$F = -\frac{GmM}{R^2}\left(1 + \frac{\sigma\pi r^2}{M} + \frac{3\sigma\pi r^4}{8MR^2} + \frac{2\lambda R}{M}\right)$$

(3)

$$\rho = R + \delta$$

$$F_{eff}(\rho) = F(R+\delta) + \frac{L^2}{m(R+\delta)^3}$$



$$= F(R) + \frac{L^2}{mR^3} + [F'(R) - \frac{3L^2}{mR^4}]\delta$$

$$= -[-F'(R) + \frac{3L^2}{mR^4}]\delta$$

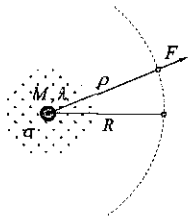
$$= -\frac{GmM}{R^3}\left(1 + \frac{\sigma\pi r^2}{M} - \frac{3\sigma\pi r^4}{8MR^2} + \frac{4\lambda R}{M}\right)\delta = -k\delta$$

$$\omega_r = \sqrt{\frac{k}{m}} = \sqrt{\frac{GM}{R^3}\left(1 + \frac{\sigma\pi r^2}{M} - \frac{3\sigma\pi r^4}{8MR^2} + \frac{4\lambda R}{M}\right)}$$

$$F = -\frac{GmM}{R^2}\left(1 + \frac{\sigma\pi r^2}{M} + \frac{3\sigma\pi r^4}{8MR^2} + \frac{2\lambda R}{M}\right), L = m\sqrt{GMR(1 + \frac{\sigma\pi r^2}{M} + \frac{3\sigma\pi r^4}{8MR^2} + \frac{2\lambda R}{M})}$$

$$\omega_0 \approx \sqrt{\frac{GM}{R^3}(1 + \frac{\sigma\pi r^2}{2M} + \frac{3\sigma\pi r^4}{16MR^2} + \frac{\lambda R}{M})}$$

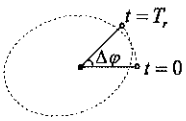
$$\omega_r \approx \sqrt{\frac{GM}{R^3}\left(1 + \frac{\sigma\pi r^2}{2M} - \frac{3\sigma\pi r^4}{16MR^2} + \frac{2\lambda R}{M}\right)}$$



$$\omega_0 T_r = 2\pi + \Delta\varphi$$

$$\Omega = \frac{|\Delta\varphi|}{T_r} = \left|\omega_0 - \frac{2\pi}{T_r}\right| = |\omega_0 - \omega_r|$$

$$= \sqrt{\frac{GM}{R^3}}\left|\frac{3\sigma\pi r^4}{8MR^2} - \frac{\lambda R}{M}\right|$$



$$\omega_0 = \sqrt{\frac{GM}{R^3}(1 + \frac{\sigma\pi r^2}{M} + \frac{3\sigma\pi r^4}{8MR^2} + \frac{2\lambda R}{M})}, \omega_r = \sqrt{\frac{GM}{R^3}\left(1 + \frac{\sigma\pi r^2}{M} - \frac{3\sigma\pi r^4}{8MR^2} + \frac{4\lambda R}{M}\right)}$$

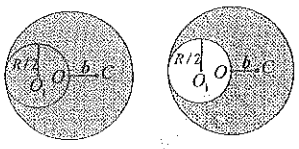
题三.

(1) $0=\dfrac{(3/4)mb+(1/4)m(-R/2)}{m}$

$b=\dfrac{1}{6}R$

$\dfrac{1}{2}mR^2+mb^2=J_C+\dfrac{1}{2}\dfrac{m}{4}(\dfrac{R}{2})^2+\dfrac{m}{4}(\dfrac{R}{2}+b)^2$

$J_C=\dfrac{37}{96}mR^2$

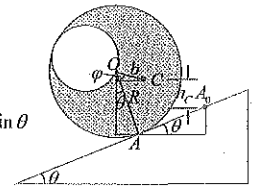


(2.1)(2.2) 假设无滑动，则系统为保守系：能量法

$E_p=\dfrac{3}{4}mgh_C=\dfrac{3}{4}mg(R\cos\theta-b\cos\varphi-R\varphi\sin\theta)$

$\dfrac{\mathrm{d}E_p}{\mathrm{d}\varphi}=\dfrac{3}{4}mg(b\sin\varphi-R\sin\theta)$

$\left.\dfrac{\mathrm{d}E_p}{\mathrm{d}\varphi}\right|_{\varphi=\varphi_0}=0\Rightarrow\sin\varphi=\dfrac{R}{b}\sin\theta=6\sin\theta$



存在平衡位置的条件：

$6\sin\theta\le 1,\ \sin\theta\le\dfrac{1}{6}\ \ (0\le\theta\le\dfrac{\pi}{2})$

以 $\varphi=0$ 时柱体与斜面的接触点 A_0 为零势能点：$\overline{AA_0}=R\varphi$

在上述条件下，平衡位置对应的 φ：

$\varphi_{01}=\arcsin(6\sin\theta),\ \varphi_{02}=\pi-\arcsin(6\sin\theta)$

$b=\dfrac{1}{6}R$

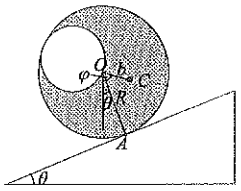
$\dfrac{\mathrm{d}^2E_p}{\mathrm{d}\varphi^2}=\dfrac{3}{4}mgb\cos\varphi$

$\varphi_{01}=\arcsin(6\sin\theta)$:

$\left.\dfrac{\mathrm{d}^2E_p}{\mathrm{d}\varphi^2}\right|_{\varphi=\varphi_{01}}>0$ ——稳定平衡

$\varphi_{02}=\pi-\arcsin(6\sin\theta)$:

$\left.\dfrac{\mathrm{d}^2E_p}{\mathrm{d}\varphi^2}\right|_{\varphi=\varphi_{01}}<0$ ——不稳定平衡



$\dfrac{\mathrm{d}E_p}{\mathrm{d}\varphi}=\dfrac{3}{4}mg(b\sin\varphi-R\sin\theta)$

$\varphi_{01}=\arcsin(6\sin\theta)$

$\varphi_{02}=\pi-\arcsin(6\sin\theta)$

处于静态平衡时，

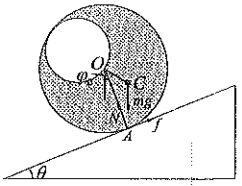
$f=mg\sin\theta$

$N=mg\cos\theta$

$f\le\mu N\Rightarrow$

$mg\sin\theta\le\mu mg\cos\theta$

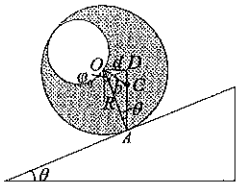
$\mu\ge\tan\theta$



另解：受力法

平衡位置：对A点合力矩为零

$\begin{cases}\sin\varphi_0=\dfrac{d}{b}\\ \sin\theta=\dfrac{d}{R}\end{cases}$



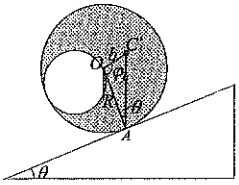
$\sin\varphi_0=\dfrac{R\sin\theta}{b}=6\sin\theta$

$\varphi_{01}=\arcsin(6\sin\theta)$

——稳定平衡位置

$\varphi_{02}=\pi-\arcsin(6\sin\theta)$

——不稳定平衡位置



处于静态平衡时，

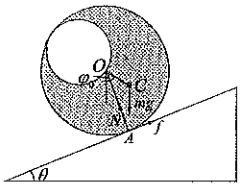
$f=mg\sin\theta$

$N=mg\cos\theta$

$f\le\mu N\Rightarrow$

$mg\sin\theta\le\mu mg\cos\theta$

$\mu\ge\tan\theta$



(2.3) 能量法：

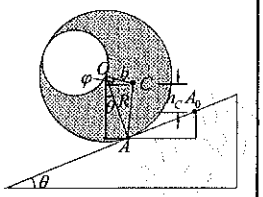
$E_p = \frac{3}{4}mg(R\cos\theta - b\cos\varphi - R\varphi\sin\theta)$

$\varphi = \varphi_0 + \delta$：

$E_p \approx E_{p0} + \frac{3}{8}mgb\cos\varphi_0\delta^2$

$= E_{p0} + \frac{1}{16}mgR\cos\varphi_0\delta^2$

$[E_{p0} = \frac{3}{4}mg(R\cos\theta - b\cos\varphi_0 - R\varphi_0\sin\theta)]$



以 $\varphi = 0$ 时柱体与斜面的接触点 A_0 为零势能参考点。

$J_A = J_C + \frac{3}{4}m[R^2 + b^2 - 2Rb\cos(\varphi - \theta)]$

$= \frac{1}{32}mR^2[37 - 8\cos(\varphi - \theta)]$

$E_k = \frac{1}{2}J_A\dot{\varphi}^2 = \frac{1}{64}mR^2[37 - 8\cos(\varphi - \theta)]\dot{\delta}^2$

$\approx \frac{1}{64}mR^2[37 - 8\cos(\varphi_0 - \theta)]\dot{\delta}^2$

$b = \frac{1}{6}R$

$J_C = \frac{37}{96}mR^2$

$\sin\varphi_0 = \frac{R\sin\theta}{b}$

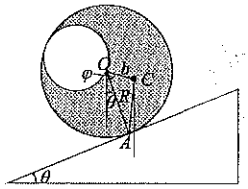
$E = E_k + E_p$

$= E_{p0} + \frac{1}{16}mgR\cos\varphi_0\delta^2 + \frac{1}{64}mR^2[37 - 8\cos(\varphi_0 - \theta)]\dot{\delta}^2$

$\frac{dE}{dt} = 0 \Rightarrow \frac{1}{8}mgR\cos\varphi_0\delta\dot{\delta} + \frac{1}{32}mR^2[37 - 8\cos(\varphi_0 - \theta)]\dot{\delta}\ddot{\delta} = 0$

$\ddot{\delta} + \frac{4\cos\varphi_0}{37 - 8\cos(\varphi_0 - \theta)}\frac{g}{R}\delta = 0$

$\omega = \sqrt{\frac{4\cos\varphi_0}{37 - 8\cos(\varphi_0 - \theta)}\frac{g}{R}}$



$E_p = E_{p0} + \frac{1}{16}mgR\cos\varphi_0\delta^2,\ E_k = \frac{1}{64}mR^2[37 - 8\cos(\varphi_0 - \theta)]\dot{\delta}^2$

另解：受力法：

研究柱体绕瞬心A的转动。A为非惯性系。

$\varphi = \varphi_0 + \delta$

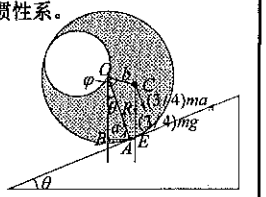
$\tau_A = -\frac{3}{4}mg(b\sin\varphi - R\sin\theta) + \tau_{惯}$

$\tau_{惯} = -\frac{3}{4}ma_A b\sin(\varphi - \theta)$

$= -\frac{3}{4}mRb\sin(\varphi - \theta)\dot{\delta}^2 \approx 0$

$(a_A = a_{An} = R\dot{\varphi}^2 = R\dot{\delta}^2)$

$\tau_A \approx -\frac{3}{4}mg(b\sin\varphi - R\sin\theta) \approx -\frac{3}{4}mgb\cos\varphi_0\delta = -\frac{1}{8}mgR\cos\varphi_0\delta$



$b = \frac{1}{6}R,\ \sin\varphi_0 = \frac{R\sin\theta}{b}$

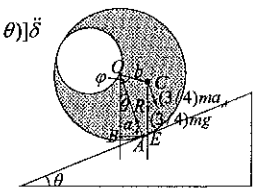
$\tau_A = J_A\beta \Rightarrow$

$-\frac{1}{8}mgR\cos\varphi_0\delta = \frac{1}{32}mR^2[37 - 8\cos(\varphi - \theta)]\ddot{\delta}$

$\approx \frac{1}{32}mR^2[37 - 8\cos(\varphi_0 - \theta)]\ddot{\delta}$

$\ddot{\delta} + \frac{4\cos\varphi_0}{37 - 8\cos(\varphi_0 - \theta)}\frac{g}{R}\delta = 0$

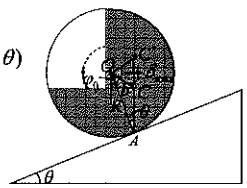
$\omega = \sqrt{\frac{4\cos\varphi_0}{37 - 8\cos(\varphi_0 - \theta)}\frac{g}{R}}$



$\tau_A \approx -\frac{1}{8}mgR\cos\varphi_0\delta,\ J_A = \frac{1}{32}mR^2[37 - 8\cos(\varphi - \theta)]$

(3)

$\delta_{max} = \varphi_{20} - \varphi_{10} = \pi - 2\arcsin(6\sin\theta)$



向下滚动允许转过的最大角度：

$\delta_{max,1} = \pi - 2\arcsin(6\sin\theta)$

向上滚动允许转过的最大角度：

$\delta_{max,2} = 2\pi - \delta_{max} = \pi + 2\arcsin(6\sin\theta)$

允许的最大增幅：

$\begin{cases}\delta_0 \le \delta_{max,1} \\ \delta_0 \le \delta_{max,2}\end{cases} \Rightarrow \delta_0 \le \delta_{max,1} = \pi - 2\arcsin(6\sin\theta)$

$\sin\theta \le \frac{1}{6}\cos\frac{\delta_0}{2}$

$\varphi_{01} = \arcsin(6\sin\theta)$

$\varphi_{02} = \pi - \arcsin(6\sin\theta)$

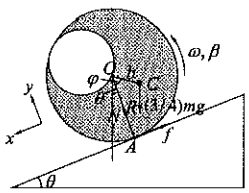
$\frac{3}{4}mg\sin\theta - f = \frac{3}{4}ma_{Cx}$

$N - \frac{3}{4}mg\cos\theta = \frac{3}{4}ma_{Cy}$

$f = \frac{3}{4}m(g\sin\theta - a_{Cx})$

$N = \frac{3}{4}m(g\cos\theta + a_{Cy})$

$a_{Cx} = ?\ a_{Cy} = ?$



$$\vec{a}_C = \vec{a}_O + \vec{a}'_C$$

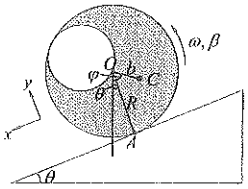
$$a_O = R\beta,\ a'_{Ct} = b\beta,\ a'_{Cn} = b\omega^2$$

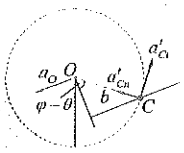
$$a_{Cx} = a_O - a'_{Ct}\cos(\varphi-\theta) + a'_{Cn}\sin(\varphi-\theta)$$

$$= [R - b\cos(\varphi-\theta)]\beta + b\sin(\varphi-\theta)\omega^2 \approx [R - b\cos(\varphi_0-\theta)]\beta$$

$$a_{Cy} = a'_{Ct}\sin(\varphi-\theta) + a'_{Cn}\cos(\varphi-\theta)$$

$$= b\sin(\varphi-\theta)\beta + b\cos(\varphi-\theta)\omega^2 \approx b\sin(\varphi_0-\theta)\beta$$





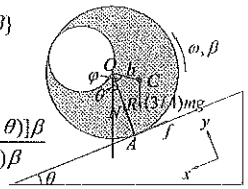
$$f = \frac{3}{4}m\{g\sin\theta - [R - b\cos(\varphi_0-\theta)]\beta\}$$

$$N = \frac{3}{4}m[g\cos\theta + b\sin(\varphi_0-\theta)\beta]$$

$$\tan\alpha = \frac{f}{N} = \frac{g\sin\theta - [R - b\cos(\varphi_0-\theta)]\beta}{g\cos\theta + b\sin(\varphi_0-\theta)\beta}$$

$$\approx \tan\theta\left(1 + \frac{b\cos\varphi_0 - R\cos\theta}{g\sin\theta\cos\theta}\beta\right)$$

$$\approx \tan\theta\left(1 - \frac{\cos\varphi_0 - 6\cos\theta}{6\sin\theta\cos\theta}\ \frac{4\cos\varphi_0}{37 - 8\cos(\varphi_0-\theta)}\delta\right)$$



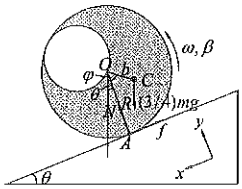
$$f = \frac{3}{4}m(g\sin\theta - a_{Cx}),\ N = \frac{3}{4}m(g\cos\theta + a_{Cy})\qquad \ddot{\delta} = -\frac{4\cos\varphi_0}{37 - 8\cos(\varphi_0-\theta)}\frac{g}{R}\delta$$

$$a_{Cx} \approx [R - b\cos(\varphi_0-\theta)]\beta,\ a_{Cy} \approx b\sin(\varphi_0-\theta)\beta\qquad b = \frac{R}{6},\ \sin\varphi_0 = \frac{R\sin\theta}{b}$$

$$\mu_{\min} = \tan\alpha_{\max}$$

$$= \tan\theta\left(1 + \frac{\cos\varphi_0 - 6\cos\theta}{6\sin\theta\cos\theta}\ \frac{4\cos\varphi_0}{37 - 8\cos(\varphi_0-\theta)}\delta_0\right)$$

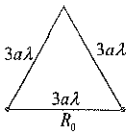
$$\left(\sin\varphi_0 = \frac{R\sin\theta}{b} \Rightarrow b\cos\varphi_0 - R\cos\theta > 0\right)$$

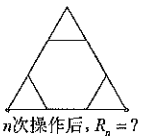


$$\tan\alpha = \tan\theta\left(1 - \frac{\cos\varphi_0 - 6\cos\theta}{6\sin\theta\cos\theta}\ \frac{4\cos\varphi_0}{37 - 8\cos(\varphi_0-\theta)}\delta\right),\ \sin\varphi_0 = \frac{R\sin\theta}{b}$$

题四.

(1) $R_0 = 2a\lambda$





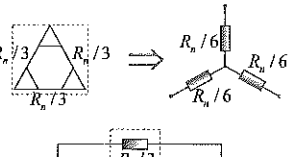
n次操作后，$R_n = ?$

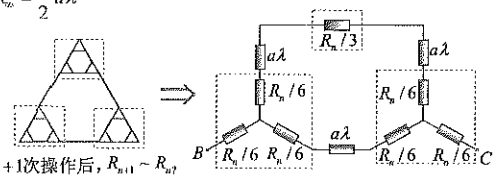
$$R_{n+1} = \cdots = \frac{5R_n + 6a\lambda}{9}$$

$$n \to \infty,\ R_{n+1} = R_n \equiv R_\infty$$

$$R_\infty = \frac{5R_\infty + 6a\lambda}{9}$$

$$R_\infty = \frac{3}{2}a\lambda$$





$n+1$次操作后，$R_{n+1} \sim R_n$?

(2)

A、E与右边断开时，对称性⇒

$$\left.\begin{aligned} V_{EA} &= V_{AC} = V_{CE} \\ V_{EA} + V_{AC} + V_{CE} &= 0 \end{aligned}\right\} \Rightarrow V_{EA} = 0$$

$$\varepsilon_1 = V_{EA} = 0,\ r_1 = R_{EA} = \frac{1}{3}R_\infty = \frac{1}{2}a\lambda$$

ΔFGD中各网孔电动势为零；

F、G与左边断开时，各网孔电流为零；

各支路电流为零。

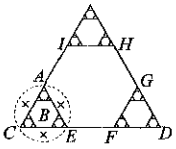
$$\varepsilon_2 = V_{FG} = \varepsilon_{FG},\ r_2 = R_{FG} = a\lambda/2$$

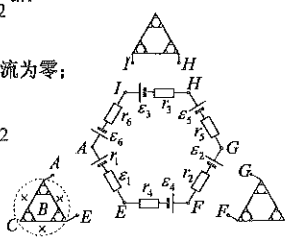
同理：$\varepsilon_3 = \varepsilon_{HI},\ r_3 = a\lambda/2$

$$\varepsilon_4 = \varepsilon_{EP},\ r_4 = a\lambda$$

$$\varepsilon_5 = \varepsilon_{GH},\ r_5 = a\lambda$$

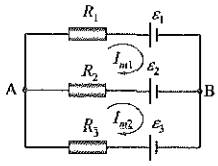
$$\varepsilon_6 = \varepsilon_{IA},\ r_6 = a\lambda$$





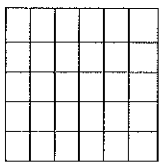
$$R_\infty = \frac{3}{2}a\lambda$$

$$I_{m1}R_1 + (I_{m1} - I_{m2})R_2 = \varepsilon_2 - \varepsilon_1$$



一般地：

$$\begin{cases} a_{11}I_{m1} + a_{12}I_{m2} + \cdots a_{1N}I_{mN} = \sum\varepsilon_1 \\ a_{21}I_{m1} + a_{22}I_{m2} + \cdots a_{2N}I_{mN} = \sum\varepsilon_2 \\ \cdots\cdots \\ a_{N1}I_{m1} + a_{N2}I_{m2} + \cdots a_{NN}I_{mN} = \sum\varepsilon_N \end{cases}$$



若$\sum\varepsilon_1 = \sum\varepsilon_2 = \cdots = \sum\varepsilon_N = 0$

若$I_{m1} = I_{m2} = \cdots = I_{mN} = 0$是方程组的一个解

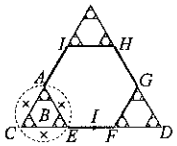
电路唯一性原理$\Rightarrow I_{m1} = I_{m2} = \cdots = I_{mN} = 0$是方程组唯一的解

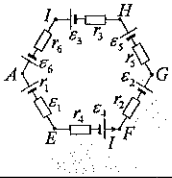
$$\Phi_{圆}=B\cdot\pi(\frac{a}{\sqrt{3}})^2=\frac{\pi}{3}kta^2$$

$$\varepsilon_{EF}+\varepsilon_{FG}+\varepsilon_{GH}+\varepsilon_{HI}+\varepsilon_{IA}+\varepsilon_{\widehat{AE}}=0$$

$$\varepsilon_{EF}+\varepsilon_{FG}+\varepsilon_{GH}+\varepsilon_{HI}+\varepsilon_{IA}=-\varepsilon_{\widehat{AE}}=\varepsilon_{\widehat{EA}}=\frac{1}{3}\frac{\mathrm{d}\Phi_{圆}}{\mathrm{d}t}=\frac{\pi}{9}ka^2$$

$$I=\frac{\varepsilon_1+\varepsilon_2\cdots+\varepsilon_6}{r_1+r_2\cdots+r_6}=\frac{\varepsilon_{EF}+\varepsilon_{FG}+\varepsilon_{GH}+\varepsilon_{HI}+\varepsilon_{IA}}{9\lambda a/2}=\frac{2\pi}{81}\frac{ka}{\lambda}$$





$\varepsilon_1=0, r_1=a\lambda/2$

$\varepsilon_2=\varepsilon_{FG}, r_2=a\lambda/2$

$\varepsilon_3=\varepsilon_{HI}, r_3=a\lambda/2$

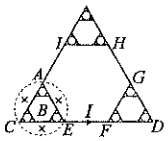
$\varepsilon_4=\varepsilon_{EF}, r_4=a\lambda$

$\varepsilon_5=\varepsilon_{GH}, r_5=a\lambda$

$\varepsilon_6=\varepsilon_{IA}, r_6=a\lambda$

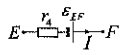
(3)

$$\begin{cases}\varepsilon_{EF}+\varepsilon_{FO}+\varepsilon_{OE}=\frac{\mathrm{d}}{\mathrm{d}t}(\frac{\varphi}{2\pi}\Phi_{圆})=\frac{\varphi}{6}ka^2\\ \varepsilon_{FO}+\varepsilon_{OE}=0\end{cases}$$

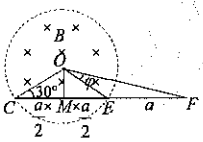


$$\varepsilon_{EF}=\frac{\varphi}{6}ka^2$$

$$\tan\angle MOF=\frac{\overline{MF}}{\overline{OM}}=\frac{3a/2}{(a/2)\tan 30^\circ}=3\sqrt{3}$$



$$\angle MOF=\arctan(3\sqrt{3})$$



$$\varphi=\angle MOF-\angle MOE=\arctan(3\sqrt{3})-\pi/3$$

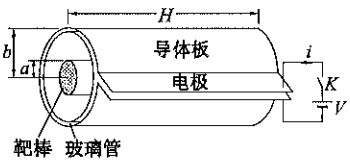
$$\varepsilon_{EF}=[\frac{1}{6}\arctan(3\sqrt{3})-\frac{\pi}{18}]ka^2$$

$$V_{EF}=Ir_4-\varepsilon_{EF}=[\frac{13\pi}{162}-\frac{1}{6}\arctan(3\sqrt{3})]ka^2\approx 0.022ka^2$$

$$I=\frac{2\pi}{81}\frac{ka}{\lambda}, r_4=a\lambda, \Phi_{圆}=\frac{\pi}{3}kta^2$$

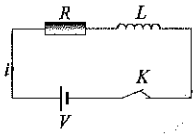
题五.

(1) $R=\frac{1}{\gamma}\frac{2\pi b}{Hd}$



$$\Phi=\mu_0 j\cdot\pi b^2=\frac{\mu_0 i}{H}\pi b^2$$

$$L=\frac{\Phi}{i}=\frac{\pi\mu_0 b^2}{H}$$



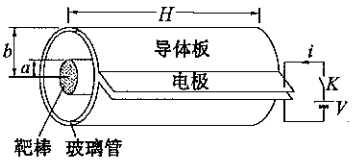
$$iR=V-L\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$\frac{\mathrm{d}i}{i-V/R}=-\frac{R}{L}\mathrm{d}t,\quad \int_0^i\frac{\mathrm{d}i}{i-V/R}=-\frac{R}{L}\int_0^t\mathrm{d}t$$

$$i=\frac{V}{R}(1-e^{-\frac{R}{L}t})=\frac{\gamma dHV}{2\pi b}(1-e^{-\frac{2t}{\mu_0\gamma bd}})$$

(2)

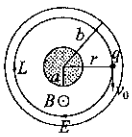
$$B=\mu_0 j=\frac{\mu_0 i}{H}$$



$$\int_L\vec{E}\cdot\mathrm{d}l=2\pi rE$$

$$\Phi=-B\cdot\pi r^2=-\frac{\pi\mu_0 r^2 i}{H}$$

$$\int_L\vec{E}\cdot\mathrm{d}l=-\frac{\mathrm{d}\Phi}{\mathrm{d}t}\Rightarrow E=\frac{\mu_0 r}{2H}\frac{\mathrm{d}i}{\mathrm{d}t}$$



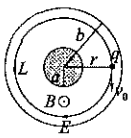
$$F=qE=\frac{\mu_0 qr}{2H}\frac{\mathrm{d}i}{\mathrm{d}t}=m\frac{\mathrm{d}v}{\mathrm{d}t}$$

$$\mathrm{d}v=\frac{\mu_0 qr}{2mH}\mathrm{d}i,\quad \int_0^{v_0}\mathrm{d}v=\frac{\mu_0 qr}{2mH}\int_0^{i_s}\mathrm{d}i\quad (r\approx 常量)$$

$$v_0=\frac{\mu_0 i_s qr}{2mH}$$

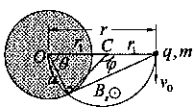
(2)

$$B_s=\frac{\mu_0 i_s}{H}$$



$$T=\frac{2\pi m}{qB_s}=\frac{2\pi mH}{\mu_0 i_s q}$$

$$r_1=\frac{mv_0}{qB_s}=\frac{1}{2}r$$



$$\cos\theta=\frac{a}{r}$$

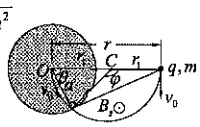
$$\varphi=2\theta=2\arccos\frac{a}{r}$$

$$t=\frac{\varphi}{2\pi}T=\frac{1}{\omega}\arccos\frac{a}{r}\quad(\omega\equiv\frac{\mu_0 i_s q}{2mH})$$

$$B=\frac{\mu_0 i}{H}, v_0=\frac{\mu_0 i_s qr}{2mH}, i_s=\frac{\gamma dHV}{2\pi b}$$

(3)

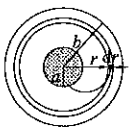
$$v_r=v_0\cos(\frac{\pi}{2}-\theta)=v_0\sin\theta=\frac{\mu_0 i_s q}{2mH}\sqrt{r^2-a^2}$$



$$\mathrm{d}p_r=mv_r\cdot n\cdot 2\pi r\mathrm{d}rH$$

$$=\pi n\mu_0 i_s q\sqrt{r^2-a^2}r\mathrm{d}r$$

$$t=\frac{1}{\omega}\arccos\frac{a}{r}\Rightarrow r=\frac{a}{\cos\omega t}$$



$$\mathrm{d}p_r=\pi n\mu_0 i_s qa^3\omega(\frac{\sin\omega t}{\cos^2\omega t})^2\mathrm{d}t$$

$$P=\frac{\mathrm{d}p_r}{\mathrm{d}t}\frac{1}{2\pi aH}=\frac{n\mu_0 i_s qa^2\omega}{2H}(\frac{\sin\omega t}{\cos^2\omega t})^2$$

$$=\frac{n(\mu_0 i_s qa)^2}{4mH^2}(\frac{\sin\omega t}{\cos^2\omega t})^2$$

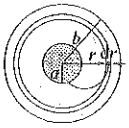
$$\cos\theta=\frac{a}{r}, v_0=\frac{\mu_0 i_s qr}{2mH}$$

$$t=\frac{2}{\omega}\arccos\frac{a}{r}, \omega\equiv\frac{\mu_0 i_s q}{2mH}$$

$$P_{总}=P_e+P_p$$

$$=\frac{n(\mu_0 i_s ea)^2}{4m_e H^2}\left[\frac{1}{m_e}\left(\frac{\sin\omega_e t}{\cos^2\omega_e t}\right)^2+\frac{1}{m_p}\left(\frac{\sin\omega_p t}{\cos^2\omega_p t}\right)^2\right]$$

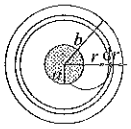
$$\left(\omega_e=\frac{\mu_0 i_s e}{2m_e H},\ \omega_p=\frac{\mu_0 i_s e}{2m_p H}\right)$$



$$P=\frac{n(\mu_0 i_s qa)^2}{4mH^2}\left(\frac{\sin\omega t}{\cos^2\omega t}\right)^2,\ \omega=\frac{\mu_0 i_s q}{2mH}$$

(4)

$$\cos\omega t=\frac{a}{r},\ \sin\omega t=\sqrt{1-\frac{a^2}{r^2}}$$



$$P=\frac{n(\mu_0 i_s er)^2}{4mH^2}\left(\frac{r^2}{a^2}-1\right)$$

$$P_{总}=P_e+P_p=\frac{n(\mu_0 i_s er)^2}{4H^2}\left(\frac{1}{m_e}+\frac{1}{m_p}\right)\left(\frac{r^2}{a^2}-1\right)\approx\frac{n(\mu_0 i_s er)^2}{4m_e H^2}\left(\frac{r^2}{a^2}-1\right)$$

$$(m_p>>m_e)$$

$$P_{max}=\frac{n(\mu_0 i_s eb)^2}{4m_e H^2}\left(\frac{b^2}{a^2}-1\right)$$

$$w_B=\frac{B_s^2}{2\mu_0}=\frac{\mu_0 i_s^2}{2H^2}$$

$$\beta=\frac{P_{max}}{w_B}=\frac{\mu_0 ne^2 b^2}{2m_e}\left(\frac{b^2}{a^2}-1\right)\approx 1.77>1$$

角向箍缩比托卡马克装置的能量利用率更高。

$$P=\frac{n(\mu_0 i_s ea)^2}{4mH^2}\left(\frac{\sin\omega t}{\cos^2\omega t}\right)^2$$

$$r=\frac{a}{\cos\omega t},\ \omega=\frac{\mu_0 i_s e}{2mH}$$

$$B_s=\frac{\mu_0 i_s}{H}$$

题六.

(1) $mg=p_0A$

$$p(l_0+x)^\gamma=p_0 l_0^\gamma\Rightarrow$$

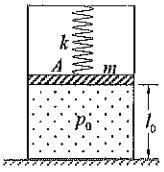
$$p=p_0\left(1+\frac{x}{l_0}\right)^{-\gamma}\approx p_0\left(1-\gamma\frac{x}{l_0}\right)$$

$$k=\frac{mg}{l_0}$$

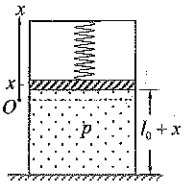
$$\sum F=pA-mg-kx=-\frac{(\gamma+1)mg}{l_0}x=-\xi x$$

$$T=2\pi\sqrt{\frac{m}{\xi}}=2\pi\sqrt{\frac{l_0}{(\gamma+1)g}}$$

$$\gamma=\frac{5}{3}\Rightarrow T=\pi\sqrt{\frac{3l_0}{2g}}$$



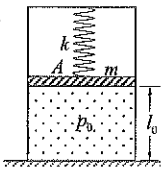
$k=mg/l_0$



(2.1)

$$p_0 l_0^\gamma=p_1 l_1^\gamma=p_2 l_2^\gamma\Rightarrow p_1=p_0\left(\frac{l_0}{l_1}\right)^\gamma,\ p_2=p_0\left(\frac{l_0}{l_2}\right)^\gamma$$

$$\begin{cases}W=\frac{1}{2}k(l_2-l_0)^2-\frac{1}{2}k(l_1-l_0)^2+mg(l_2-l_1)\\ W=\frac{(p_1l_1-p_2l_2)A}{\gamma-1}=\frac{p_0Al_0}{\gamma-1}\left[\left(\frac{l_0}{l_1}\right)^{\gamma-1}-\left(\frac{l_0}{l_2}\right)^{\gamma-1}\right]\\ p_0A=mg,\ k=mg/l_0\end{cases}$$



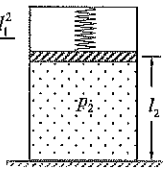
$k=mg/l_0$

$$\frac{1}{\gamma-1}\left[\left(\frac{l_0}{l_1}\right)^{\gamma-1}-\left(\frac{l_0}{l_2}\right)^{\gamma-1}\right]=\frac{l_2^2-l_1^2}{2l_0^2}$$

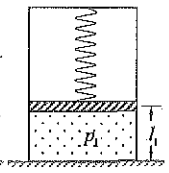
$$\gamma=5/3,\ l_1=l_0/2\Rightarrow$$

$$\left(\frac{l_0}{l_2}\right)^{2/3}+\frac{1}{3}\left(\frac{l_2}{l_0}\right)^2=2^{2/3}+\frac{1}{12}$$

$$l_2\approx 1.71l_0,\ h_m=l_2-l_1=1.21l_0$$



活塞升至最高时



活塞刚释放时

$l_1=l_0/2$

(2.2)

被提起的条件：$k\Delta l=p_3A$

$$k=mg/l_0$$

$$\Delta l=l_3-l_0$$

$$p_3 l_3^\gamma=p_0 l_0^\gamma\Rightarrow p_3=p_0\left(\frac{l_0}{l_3}\right)^\gamma$$

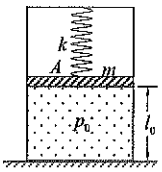
$$\frac{mg}{l_0}(l_3-l_0)=p_0A\left(\frac{l_0}{l_3}\right)^\gamma$$

$$\frac{l_3}{l_0}-1=\left(\frac{l_0}{l_3}\right)^\gamma=\left(\frac{l_0}{l_3}\right)^{5/3}$$

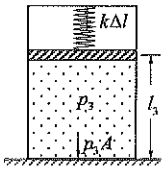
$$l_3\approx 1.506l_0<l_2$$

汽缸能被提起。

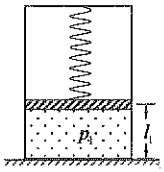
$$l_2\approx 1.71l_0,\ mg=p_0A$$



$k=mg/l_0$



汽缸被提起时



活塞刚释放时

$l_1=l_0/2$

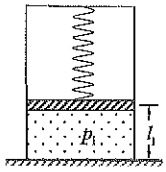
$$\begin{cases}W=\frac{1}{2}k(l_3-l_0)^2-\frac{1}{2}k(l_1-l_0)^2+mg(l_3-l_1)+\frac{1}{2}mv^2\\ W=\frac{(p_1l_1-p_3l_3)A}{\gamma-1}=\frac{p_0Al_0}{\gamma-1}\left[\left(\frac{l_0}{l_1}\right)^{\gamma-1}-\left(\frac{l_0}{l_3}\right)^{\gamma-1}\right]\\ p_0A=mg,\ k=mg/l_0\end{cases}$$

$$v^2=2gl_0\left[\frac{(l_0/l_1)^{\gamma-1}-(l_0/l_3)^{\gamma-1}}{\gamma-1}-\frac{l_3^2-l_1^2}{2l_0^2}\right]$$

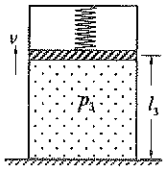
$$\gamma=\frac{5}{3},\ l_1=\frac{1}{2}l_0,\ l_3=1.506l_0$$

$$v^2\approx 0.462gl_0$$

$$p_1=p_0\left(\frac{l_0}{l_1}\right)^\gamma,\ p_3=p_0\left(\frac{l_0}{l_3}\right)^\gamma,\ p_0A=mg,\ l_3\approx 1.506l_0$$



活塞刚释放时



汽缸被提起时

因为汽缸质量可忽略不计，汽缸所受的合力恒为零。

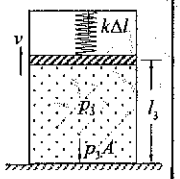
汽缸被提起后，$k\Delta l = p_3 A$ 恒成立。$\Rightarrow l_3$不变。

汽缸被提起后，活塞相对汽缸的速度为零。

汽缸与活塞共同以初速度v，加速度$-g$做竖直上抛运动。

汽缸被提起最大高度：

$$H_m = \frac{v^2}{2g} \approx 0.23 l_0$$



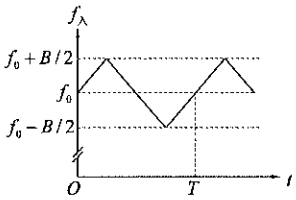
汽缸被提起时

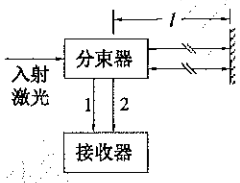
$v^2 \approx 0.462 g l_0$

题七.

(1)

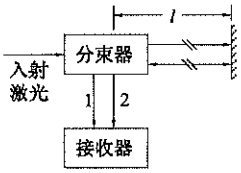
$$f_{入}(t) = \begin{cases} \frac{2B}{T}t + f_0, 0 \le t \le \frac{1}{4}T \\ -\frac{2B}{T}t + f_0 + B, \ \frac{1}{4}T \le t \le \frac{3}{4}T \\ \frac{2B}{T}t + f_0 - 2B, \ \frac{3}{4}T \le t \le T \end{cases}$$





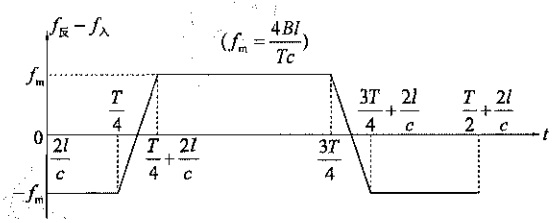
$$f_{反}(t) = f_{入}(t - \frac{2l}{c})$$

$$= \begin{cases} \frac{2B}{T}t - \frac{2l}{c} + f_0, \frac{2l}{c} \le t \le \frac{1}{4}T + \frac{2l}{c} \\ -\frac{2B}{T}(t - \frac{2l}{c}) + f_0 + B, \ \frac{1}{4}T + \frac{2l}{c} \le t \le \frac{3}{4}T + \frac{2l}{c} \\ \frac{2B}{T}(t - \frac{2l}{c}) + f_0 - 2B, \ \frac{3}{4}T + \frac{2l}{c} \le t \le T + \frac{2l}{c} \end{cases}$$

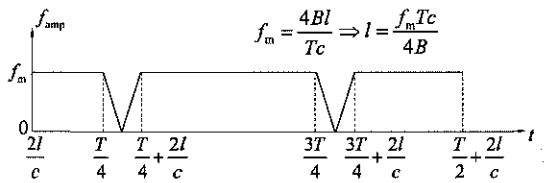


$$f_{反}(t) - f_{入}(t) = \begin{cases} -\frac{4Bl}{Tc}, \ \frac{2l}{c} \le t \le \frac{1}{4}T \\ \frac{4B}{T}t - B - \frac{4Bl}{Tc}, \ \frac{1}{4}T \le t \le \frac{1}{4}T + \frac{2l}{c} \\ \frac{4Bl}{Tc}, \ \frac{1}{4}T + \frac{2l}{c} \le t \le \frac{3}{4}T \\ -\frac{4B}{T}t + 3B + \frac{4Bl}{Tc}, \ \frac{3}{4}T \le t \le \frac{3}{4}T + \frac{2l}{c} \\ -\frac{4Bl}{Tc}, \ \frac{3}{4}T + \frac{2l}{c} \le t \le T + \frac{2l}{c} \end{cases}$$

$$f_{入}(t) = \begin{cases} \frac{2B}{T}t + f_0, 0 \le t \le \frac{1}{4}T \\ -\frac{2B}{T}t + f_0 + B, \ \frac{1}{4}T \le t \le \frac{3}{4}T \\ \frac{2B}{T}t + f_0 - 2B, \ \frac{3}{4}T \le t \le T \end{cases}, \quad f_{反}(t) = \begin{cases} \frac{2B}{T}t - \frac{2l}{c} + f_0, \frac{2l}{c} \le t \le \frac{1}{4}T + \frac{2l}{c} \\ -\frac{2B}{T}(t - \frac{2l}{c}) + f_0 + B, \ \frac{1}{4}T + \frac{2l}{c} \le t \le \frac{3}{4}T + \frac{2l}{c} \\ \frac{2B}{T}(t - \frac{2l}{c}) + f_0 - 2B, \ \frac{3}{4}T + \frac{2l}{c} \le t \le T + \frac{2l}{c} \end{cases}$$



$$f_{amp}(t) = |f_{反}(t) - f_{入}(t)|$$

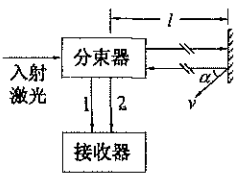


(2)

地面：入射到汽车的激光频率：f

汽车：接收或反射的激光频率：

$$f' = \frac{1 - \beta\cos(\pi + \alpha)}{\sqrt{1 - \beta^2}} f \quad (\beta = \frac{v}{c})$$



地面：汽车反射的的激光频率：

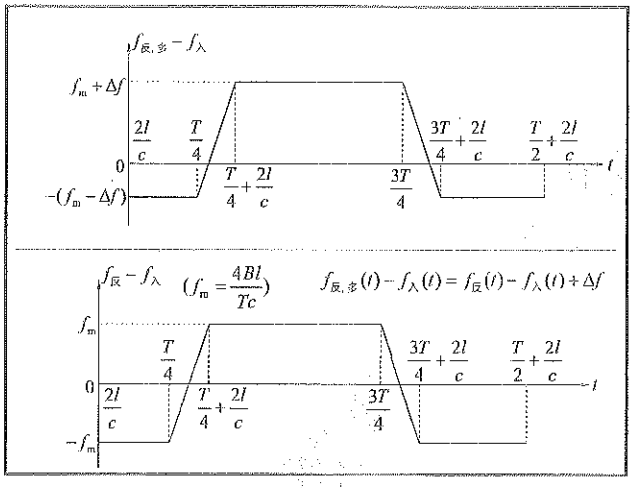
$$f_{反,多} = \frac{\sqrt{1 - \beta^2}}{1 - \beta\cos\alpha} f' = \frac{1 + \beta\cos\alpha}{1 - \beta\cos\alpha} f \approx (1 + 2\beta\cos\alpha) f$$

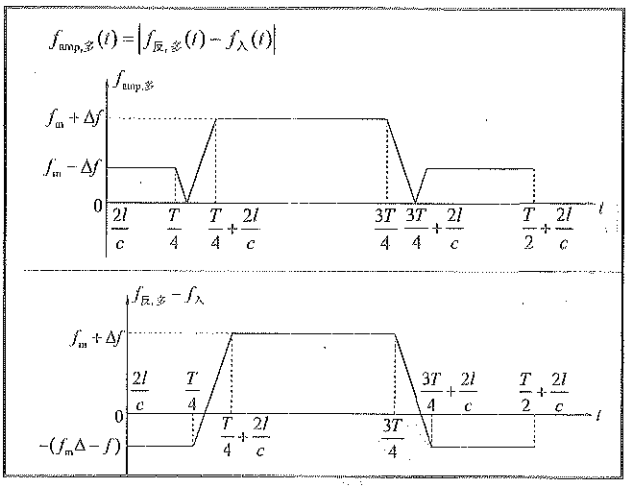
$$f_{反,多}(t) = (1 + 2\beta\cos\alpha) f_{入}(t - \frac{2l}{c}) \approx f_{入}(t - \frac{2l}{c}) + 2\beta f_0 \cos\alpha$$

$$= f_{反}(t) + \Delta f \quad (\Delta f \equiv 2\beta f_0 \cos\alpha)$$

$$f_{反,多}(t) - f_{入}(t) = f_{反}(t) - f_{入}(t) + \Delta f$$

$$\nu' = \frac{\nu(1 - \beta\cos\theta)}{\sqrt{1 - \beta^2}} = \frac{\nu\sqrt{1 - \beta^2}}{1 + \beta\cos\theta'}, \nu = \frac{\nu'(1 + \beta\cos\theta')}{\sqrt{1 - \beta^2}} = \frac{\nu'\sqrt{1 - \beta^2}}{1 - \beta\cos\theta}$$

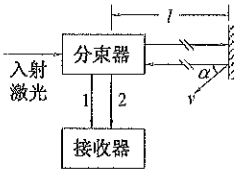


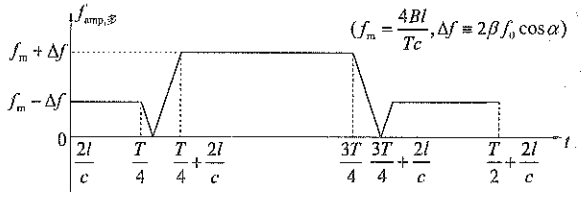


$$f'_m=f_m+\Delta f\quad=f_m+2\beta f_0\cos\alpha$$

$$=f_m+\frac{2v}{c}f_0\cos\alpha$$

$$v=\frac{f'_m-f_m}{2f_0\cos\alpha}c\quad(f_m=\frac{4Bl}{Tc})$$





题八.

(1) $S:(l_0,0)$

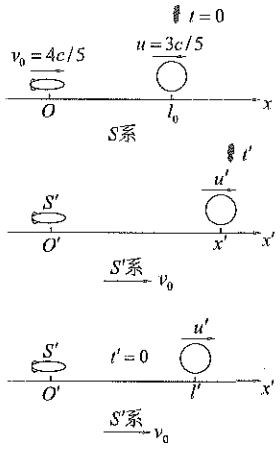
$S':(x',t')$

$$t'=\frac{0-l_0v_0/c^2}{\sqrt{1-v_0^2/c^2}}=-\frac{4}{3}\frac{l_0}{c}$$

$$x'=\frac{l_0-v_0\cdot 0}{\sqrt{1-v_0^2/c^2}}=\frac{5}{3}l_0$$

$$u'=\frac{u-v_0}{1-uv_0/c^2}=-\frac{5}{13}c$$

$$l'=x'-u't'=\frac{15}{13}l_0$$



(2)

$$\begin{cases}dp'=-dp'_\gamma=-\frac{wdt'}{c}\\ dp'=\frac{E'_0+dE'}{c^2}dv'\approx\frac{E'_0}{c^2}dv'=m_0dv'\end{cases}$$

$$\frac{dv'}{dt'}=-\frac{w}{m_0c}$$

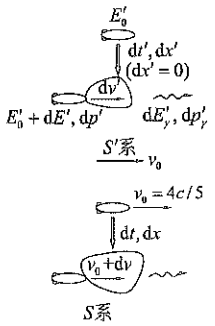
$$v_0+dv=\frac{dv'+v_0}{1+v_0dv'/c^2}\approx v_0+(1-\frac{v_0^2}{c^2})dv'$$

$$\Rightarrow dv=(1-v_0^2/c^2)dv'$$

$$dt=\frac{dt'+dx'v_0/c^2}{\sqrt{1-v_0^2/c^2}}=\frac{dt'}{\sqrt{1-v_0^2/c^2}}$$

$$\Rightarrow dt'=dt\sqrt{1-v_0^2/c^2}$$

$$a_0=\frac{dv}{dt}=\frac{dv'}{dt'}(1-\frac{v_0^2}{c^2})^{3/2}=-\frac{27}{125}\frac{w}{m_0c}$$



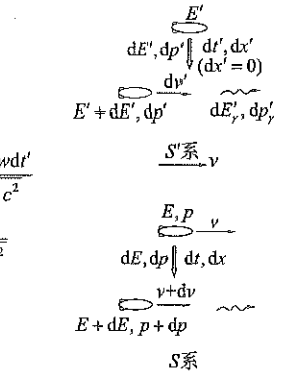
(3)

$$dE'=-dE'_\gamma=-wdt'$$

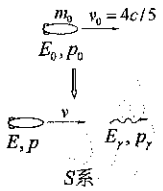
$$dp'=-dp'_\gamma=-\frac{w}{c}dt'$$

$$\begin{cases}dE=\frac{dE'+vdp'}{\sqrt{1-v^2/c^2}}=-\frac{(1+v/c)wdt'}{\sqrt{1-v^2/c^2}}\\ dt=\frac{dt'+vdx'/c^2}{\sqrt{1-v^2/c^2}}=\frac{dt'}{\sqrt{1-v^2/c^2}}\end{cases}$$

$$dE=-(1+\frac{v}{c})wdt\quad(1)$$



$$\begin{cases} E+E_\gamma=E_0=\dfrac{m_0c^2}{\sqrt{1-v_0^2/c^2}}=\dfrac{5}{3}m_0c^2 \qquad E\sim v? \\ p+p_\gamma=p_0=\dfrac{E_0}{c}v_0=\dfrac{4}{3}m_0c \\ E_\gamma=p_\gamma c \\ p=\dfrac{E}{c^2}v \end{cases}$$



$$E=\frac{1}{3}\frac{m_0c^2}{1-v/c},\quad \mathrm{d}E=\frac{1}{3}\frac{m_0c}{(1-v/c)^2}\mathrm{d}v \quad (2)$$

$$(1)(2)\Rightarrow\ -(1+\frac{v}{c})w\mathrm{d}t=\frac{1}{3}\frac{m_0c}{(1-v/c)^2}\mathrm{d}v$$

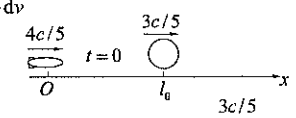
$$\mathrm{d}t=-\frac{m_0c}{3w}\frac{1}{(1-v/c)^2(1+v/c)}\mathrm{d}v$$

$$\mathrm{d}x=v\mathrm{d}t=-\frac{m_0c}{3w}\frac{v}{(1-v/c)^2(1+v/c)}\mathrm{d}v$$

$$\mathrm{d}E=-(1+\frac{v}{c})w\mathrm{d}t \quad (1)$$

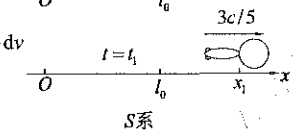
$$t_1=-\frac{m_0c}{3w}\int_{4c/5}^{3c/5}\frac{1}{(1-v/c)^2(1+v/c)}\mathrm{d}v$$

$$=\frac{m_0c^2}{12w}(5+2\ln\frac{3}{2})$$



$$x_1=-\frac{m_0c}{3w}\int_{4c/5}^{3c/5}\frac{v}{(1-v/c)^2(1+v/c)}\mathrm{d}v$$

$$=\frac{m_0c^3}{12w}(5-2\ln\frac{3}{2})$$



$$x_1-\frac{3}{5}ct_1=l_0\Rightarrow\frac{m_0c^3}{12w}(5-2\ln\frac{3}{2})-\frac{3}{5}c\cdot\frac{m_0c^2}{12w}(5+2\ln\frac{3}{2})=l_0$$

$$w=\frac{m_0c^3}{l_0}(\frac{1}{6}-\frac{4}{15}\ln\frac{3}{2})$$

$$\mathrm{d}t=-\frac{m_0c}{3w}\frac{1}{(1-v/c)^2(1+v/c)}\mathrm{d}v,\mathrm{d}x=-\frac{m_0c}{3w}\frac{v}{(1-v/c)^2(1+v/c)}\mathrm{d}v$$